Discharging the anti-concentration hypothesis (AC): the modulus form, and why bounded density is the wrong one

Working note for the "Keep the Angle" program

Purpose. The rank-transfer theorem (companion reductions note, "Rank transfer without bi-Lipschitz hypotheses") assumes an anti-concentration hypothesis to control the fraction of near-tied comparisons. It was stated as *bounded density*:

$$\text{(AC)} \qquad \kappa_\Lambda := \sup_{x \in M} \big\| \text{density of } V(x,Y) \big\|_\infty < \infty, \qquad V(x,y) := \|N_\Lambda(x) - N_\Lambda(y)\|,$$

Y drawn from the sampling density. This note does two things. First, it shows that (AC) as bounded density is *generically false* in dimension 2: a nondegenerate saddle of $V(x,\cdot)$ forces a logarithmic divergence of the density. Second, it proves the *modulus* form that the transfer theorem actually consumes—and it is the correct hypothesis, exactly as the reductions note anticipated ("bounded density is the simplest sufficient form, not the needed one"). For real-analytic metrics the modulus form holds unconditionally, by a uniform Łojasiewicz sublevel estimate, so (AC) is discharged on the standard examples (tori, spheres, projective and homogeneous spaces, and their real-analytic deformations). Notation follows the reductions note; the cutoff Λ is fixed and multiplet-closed, and N_Λ is the normalized truncated commute-weighted eigenmap.

1 Bounded density is the wrong hypothesis

Fix x and consider $V(x,\cdot) : M \to [0,2]$, $V(x,y)^2 = 2-2C_\Lambda(x,y)$ with $C_\Lambda(x,y) = \langle N_\Lambda(x), N_\Lambda(y)\rangle$. At a fixed cutoff N_Λ is a smooth map, so $V(x,\cdot)$ is smooth, and its pushforward density is governed by the coarea formula

$$f_{V(x,\cdot)}(t) = \int_{\{V(x,\cdot)=t\}} \frac{\rho(y)}{|\nabla_y V(x,y)|}\, d\mathcal{H}^{d-1}(y), \tag{1}$$

ρ the sampling density. The integrand blows up where $\nabla_y V = 0$, i.e. at critical points of $V(x,\cdot)$, and whether (1) stays bounded is decided by the *type* of those critical points.

Proposition 1 (Saddles break bounded density in $d=2$)**.** *Let $d = 2$ and suppose $V(x,\cdot)$ has a nondegenerate (Morse) saddle at some $y_0 \neq x$, with $V(x,y_0) = t_0$ and $\rho(y_0) > 0$. Then $f_{V(x,\cdot)}(t) \to \infty$ as $t \to t_0$; indeed $f_{V(x,\cdot)}(t) \geq c \log \frac{1}{|t-t_0|}$ near t_0. In particular $\kappa_\Lambda = \infty$, so (AC) as bounded density fails.*

Proof. In Morse coordinates $V(x,\cdot) = t_0 + \frac{1}{2}(\xi_1^2 - \xi_2^2) + o(|\xi|^2)$ near y_0, so level sets near t_0 are hyperbolas $\xi_1^2 - \xi_2^2 = 2(t-t_0)$ and $|\nabla V| \asymp |\xi|$. Integrating $|\xi|^{-1}$ along a level hyperbola over a fixed coordinate neighborhood gives a contribution $\asymp \int \frac{d\ell}{|\xi|} \asymp \log \frac{1}{|t-t_0|}$ (the standard logarithmic saddle divergence of the density of states). With $\rho(y_0) > 0$ the weight is bounded below, so $f_{V(x,\cdot)}(t) \gtrsim \log \frac{1}{|t-t_0|}$. $\square$

Saddles are forced on positive genus: on a closed surface with Euler characteristic $\chi \le 0$ (genus ≥ 1), $\#\mathrm{min} - \#\mathrm{saddle} + \#\mathrm{max} = \chi$ with $\#\mathrm{min}, \#\mathrm{max} \ge 1$ gives $\#\mathrm{saddle} \ge 2 - \chi \ge 2$, so a Morse $V(x, \cdot)$ always has saddles and bounded density fails—in particular on the flat tori of the paper's torus experiments. (On S^2, $\chi = 2$ permits a saddle-free height-type V; and on the round sphere $V(x, \cdot) = h(d_M(x, \cdot))$ is non-Morse, with degenerate critical *circles* where bounded density can still fail through a distinct mechanism, but is not forced by Morse theory. So the failure is not universal—it is guaranteed exactly where Morse theory forces saddles, $\chi \le 0$.) Where it occurs the divergence is only logarithmic, so the *measure* of near-tied pairs is still $O(\varepsilon \log \frac{1}{\varepsilon})$; the transfer theorem, which needs only that measure, is unharmed. This is precisely why the modulus form below is the right object.

2 The modulus form, and what the transfer theorem needs

Definition 2 ((AC′), modulus form)**.** *N_Λ satisfies* (AC′) *with modulus ω if $\omega : [0, \infty) \to [0, \infty)$ is nondecreasing with $\omega(0^+) = 0$ and*

$$\sup_{x \in M} \sup_{t \in \mathbb{R}} \mathbb{P}_Y\big(|V(x, Y) - t| \le \varepsilon\big) \;\le\; \omega(\varepsilon) \qquad \text{for all } \varepsilon > 0.$$

Lemma 3 ((AC′) is what the transfer proof uses)**.** *In the proof of the rank-transfer theorem, every appearance of (AC) may be replaced by (AC′): the flip-fraction bound becomes $\mathbb{E} f_n \le \omega(2\bar{\varepsilon}_n)$ (disjoint and anchor-sharing pairs-of-pairs alike), and the conclusion holds with the error term $C\kappa_\Lambda \bar{\varepsilon}_n$ replaced by $\omega(2\bar{\varepsilon}_n)$. Consequently the rank correlations satisfy $\widehat{\tau}_K, \widehat{\rho}_S \ge \tau_\star, \rho_\star - C\,\omega(2\bar{\varepsilon}_n) - o_P(1)$, and the transfer holds whenever $\omega(2\bar{\varepsilon}_n) \to 0$.*

Proof. The only role of (AC) in the transfer proof is Step 2, which bounds, for a pair-of-pairs $\{P, Q\}$, the probability $\mathbb{P}(|V_P - V_Q| \le 2\bar{\varepsilon}_n)$. For vertex-disjoint $\{P, Q\}$, condition on $V_Q = t$ and apply the inner bound to $V_P = V(X, Y)$: averaging (1)'s tail, $\mathbb{P}(|V_P - t| \le 2\bar{\varepsilon}_n) \le \sup_{x,t} \mathbb{P}_Y(|V(x, Y) - t| \le 2\bar{\varepsilon}_n) \le \omega(2\bar{\varepsilon}_n)$ by (AC′) after further conditioning on the free endpoint of P. For anchor-sharing $P = (x_i, x_j), Q = (x_i, x_k)$, condition on x_i and V_Q: given x_i, $V_P = V(x_i, x_j)$ with x_j independent, and (AC′) at the anchor x_i gives the same bound—this is exactly the uniform-in-x (equivalently uniform-in-anchor) form. Hence $\mathbb{E} f_n \le \omega(2\bar{\varepsilon}_n)$, and Markov + the Diaconis–Graham/Spearman conversion of Step 1 carry the rest verbatim. □

3 (AC′) holds for real-analytic metrics

Theorem 4 (Uniform Łojasiewicz modulus)**.** *Let M be a closed real-analytic Riemannian manifold and Λ a fixed multiplet-closed cutoff for which N_Λ separates points (so $V(x, \cdot)$ is non-constant for every x; this holds past the immersion threshold). Assume the sampling density ρ is bounded. Then there are constants $C_\Lambda < \infty$ and $\theta_\Lambda \in (0, 1]$, depending only on (M, Λ), with*

$$\sup_{x \in M} \sup_{t \in \mathbb{R}} \operatorname{vol}\big\{y : |V(x, y) - t| \le \varepsilon\big\} \;\le\; C_\Lambda\, \varepsilon^{\theta_\Lambda} \qquad (\varepsilon > 0),$$

so (AC′) holds with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$. In particular $\omega(2\bar{\varepsilon}_n) = O(\bar{\varepsilon}_n^{\theta_\Lambda}) \to 0$, and by Lemma 3 the rank-transfer theorem holds with (AC) discharged.

Proof. On a real-analytic manifold with a real-analytic metric the Laplace eigenfunctions are real-analytic (elliptic regularity for operators with real-analytic coefficients), so the finite combinations $G_\Lambda(x,\cdot) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)\varphi_k(\cdot)$ and $S_\Lambda = G_\Lambda(\cdot,\cdot)$ are real-analytic, and hence so is

$$u_x(y) := V(x,y)^2 = 2 - \frac{2G_\Lambda(x,y)}{\sqrt{S_\Lambda(x)S_\Lambda(y)}},$$

jointly real-analytic in $(x,y) \in M \times M$ (the denominator is bounded below: $S_\Lambda > 0$ since Λ exceeds the first nonzero eigenvalue). For each fixed x, u_x is a non-constant real-analytic function on the compact M (non-constant because N_Λ separates points), so by the Łojasiewicz sublevel-set volume estimate there are $C(x), \theta(x)$ with $\mathrm{vol}\{|u_x - s| \le \delta\} \le C(x)\delta^{\theta(x)}$ uniformly in s.

For the uniformity over x we do not appeal to semicontinuity of the exponent (which is not, in general, enough to keep both $\theta(x)$ bounded below and $C(x)$ bounded above); instead we use o-minimality. Consider the joint concentration function

$$\mathcal{G}(x,\delta) := \sup_s \mathrm{vol}\{y : |u_x(y) - s| \le \delta\}.$$

Since $(x,y) \mapsto u_x(y)$ is real-analytic on the compact $M \times M$, the fiber volume of the subanalytic family $\{(x,y,s) : |u_x(y)-s| \le \delta\}$ is a log-analytic function of its parameters (Comte–Lion–Rolin), hence $\mathcal{G}$ is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$; and $\mathcal{G}(x,0) = 0$ for every x, because each non-constant real-analytic u_x has null level sets. The o-minimal Łojasiewicz inequality, applied on the compact base $M \times [0,\delta_0]$ to the definable $\mathcal{G} \ge 0$ vanishing on the definable set $\{\delta = 0\}$, yields a single $\theta_\Lambda \in (0,1]$ and $C_\Lambda < \infty$ with $\mathcal{G}(x,\delta) \le C_\Lambda\, \delta^{\theta_\Lambda}$ uniformly in x; that is, $\sup_x \sup_s \mathrm{vol}\{|u_x - s| \le \delta\} \le C_\Lambda \delta^{\theta_\Lambda}$.

Transfer to $V = \sqrt{u}$. On $\{V \ge t_1 > 0\}$ the map $u \mapsto \sqrt{u}$ is bi-Lipschitz, so $\{|V - t| \le \varepsilon\} \subseteq \{|u - t^2| \le C'\varepsilon\}$ for $t \ge t_1$ and $\mathrm{vol} \le C_\Lambda(C'\varepsilon)^{\theta_\Lambda}$. Near $V = 0$ ($t < t_1$): $V = 0$ only where $N_\Lambda(y) = N_\Lambda(x)$; since N_Λ separates points this is the single point $y = x$ (or a subanalytic set of dimension $< d$, hence measure zero), and the sublevel $\{V \le t_1\}$ has $\mathrm{vol}\{|V - t| \le \varepsilon\} \le \mathrm{vol}\{u \le (t_1 + \varepsilon)^2\} \le C_\Lambda(t_1 + \varepsilon)^{2\theta_\Lambda}$; choosing $t_1 \asymp \varepsilon^{1/2}$ absorbs this into the same power law up to a constant. Multiplying by $\|\rho\|_\infty$ converts volume to probability, giving (AC′) with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$ (after adjusting C_Λ). The transfer-theorem consequence is Lemma 3. □

Remark 5 (The exponent, and the two mitigations retained). θ_Λ is the reciprocal of the worst Łojasiewicz multiplicity of the family $\{u_x\}$; for Morse u_x one has $\theta_\Lambda = 1$ in $d \ge 3$ and $\theta_\Lambda = 1$ away from saddles with a logarithmic correction at saddle values in $d = 2$ (matching Proposition 1: the modulus is $\varepsilon \log(1/\varepsilon)$ there, still $\to 0$). Both mitigations noted in the reductions note survive and are strengthened: (i) ω is directly measurable per substrate (estimate the empirical modulus of sampled V-values), turning (AC′) into a per-instance certificate with a proven a-priori form on analytic geometries; (ii) only $\omega(2\bar{\varepsilon}_n) \to 0$ is needed, and $\omega(\varepsilon) = C\varepsilon^\theta$ delivers a definite convergence *rate* $O(\bar{\varepsilon}_n^{\theta_\Lambda})$, sharper than the qualitative statement.

Remark 6 (Growing cutoff). Theorem 4 is at fixed Λ; the constant C_Λ degrades as $\Lambda \to \infty$ because the angular distances concentrate at $\sqrt{2}$. Quantitatively the density scale is $\kappa_\Lambda \asymp A_\Lambda$ (reductions note, Remark on the growing window), i.e. $\omega_\Lambda(\varepsilon) \asymp A_\Lambda\varepsilon$ in the regime where the Morse structure is nondegenerate, so $\omega_\Lambda(2\bar{\varepsilon}_n) \asymp A_\Lambda\bar{\varepsilon}_n \asymp E_n\sqrt{A_m}$ (using $\bar{\varepsilon}_n = 4E_n/\rho_\Lambda^-$ and $\rho_\Lambda^- \asymp \sqrt{A_\Lambda}$). This reproduces the sharpened growing-window threshold $E_n\sqrt{A_m} \to 0$ a third time—now from the analytic-modulus side—consistent with the kernel-rescaling and anti-concentration-counting derivations already on record.

4 Imported facts

(1) Elliptic regularity with real-analytic coefficients: eigenfunctions of the Laplace–Beltrami operator on a real-analytic Riemannian manifold are real-analytic (Morrey–Nirenberg). (2) Łojasiewicz sublevel-set volume estimate: for a non-constant real-analytic f on a compact set, $\mathrm{vol}\{|f - s| \le \delta\} \le C\delta^{\theta}$ with $\theta \in (0, 1]$, uniformly in s over a compact range. (2′) Uniformity over a compact real-analytic family via o-minimality: fiber volumes of subanalytic families are log-analytic, hence definable in $\mathbb{R}_{\mathrm{an,exp}}$ (Comte–Lion–Rolin, *Comptes Rendus* 2000; van den Dries–Miller, o-minimality), and the o-minimal Łojasiewicz inequality gives a single exponent and constant uniformly over the compact base (van den Dries, *Tame Topology and O-minimal Structures*). (3) The coarea formula (Federer). (4) The rank-transfer theorem and its Diaconis–Graham/Hoeffding machinery (reductions note), into which (AC′) plugs by Lemma 3.

Open. A *smooth* (non-analytic) metric can in principle have a positive-measure level set of $V(x, \cdot)$ (a plateau), which would create an atom and break even (AC′); ruling this out for smooth metrics—or exhibiting it—is the residual question. It does not arise on any manifold in the paper's examples, all real-analytic. A second refinement: an effective θ_Λ in terms of the eigenvalue gaps and $\|\varphi_k\|_{C^2}$, which would make the convergence rate explicit rather than merely positive.